Design for Sheet Lamination 307

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Class Outline

Sheet Lamination and DFAM
Sheet Lamination Materials
Sheet Lamination Applications
DFAM Considerations for Sheet Lamination
Review: DFAM for Sheet Lamination
Sheet Lamination Prototyping Methods
Sheet Lamination Production Methods
Ultrasonic Additive Manufacturing
Post-Processing for Sheet Lamination
Review: Sheet Lamination Processes

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Lesson: 1/10

Sheet Lamination and DFAM

Creating parts with any additive manufacturing (AM) process involves following the same five basic steps. The first step of design for additive manufacturing (DFAM) is conceptualization, during which manufacturers determine the type of part and its functionality. The second step is planning, when they decide how to manage the workflow to produce the part. Next is designing, which involves generating a digital model and creating a prototype of the part. The next phase, testing, involves determining how the part functions and if it performs according to specifications. If the part passes testing, it is ready for the production phase.

The specific capabilities and limitations of different AM processes affect how engineers and manufacturers approach each phase of DFAM. Sheet lamination is an AM process that bonds successive layers of material to form a part based on a computer-aided design (CAD) model. The material feedstock used for sheet lamination comes in a feedstock roll installed on the sheet lamination machine. The feedstock roll is fed onto the machine's build platform in separate sheets which are bonded together. Sheet lamination can use a variety of different materials and bonding methods to create parts. The materials are cut into the shape of the part with subtractive manufacturing methods. Sheet lamination is often considered a hybrid manufacturing process because it uses both additive and subtractive methods.

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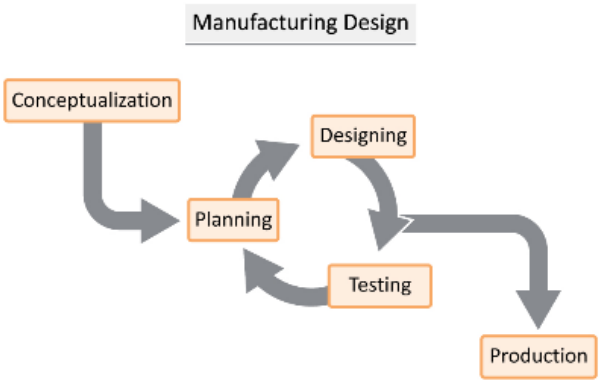

Figure 1:

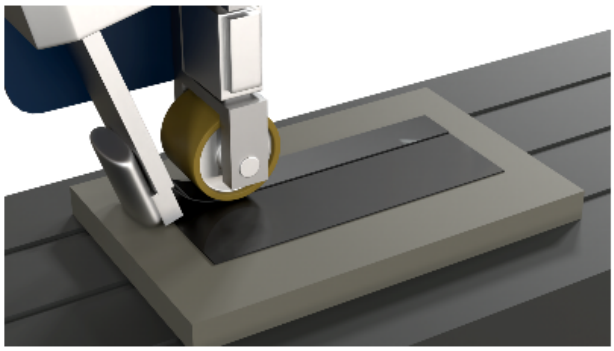

Figure 2:

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Sheet Lamination Materials

Paper Material

The first sheet lamination machines used only paper materials to build parts. Sheet lamination processes that build parts from paper materials are currently used for rapid prototyping but not for producing end-use parts.

Plastic Material

Some sheet lamination processes can use polymer materials. Polymers are plastic materials that include both thermoplastics and thermosets. Polymer materials can be used for rapid prototyping, but they are also useful for end-use products, like storage containers and medical equipment. Polymers are generally ductile materials with high corrosion- and wear-resistance. Polymer sheet lamination can use most common polymers, including polystyrene, polypropylene, nylon, and other thermoplastics.

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Composite Material

Ceramic materials, including some composite materials, have become increasingly popular as sheet lamination technology has advanced. Common ceramics used in sheet lamination include glass, cement, and ceramic matrix composites (CMCs). Ceramic materials have very high heat- and corrosion-resistance properties and are suitable for a wide variety of end-use products. Composite materials, such as carbon fiber, can also have high heat- and corrosion-resistance, as well as high strength or other properties depending on the combination of materials.

Metal Material

Metal materials are the most common sheet lamination materials for building end-use products. Sheet lamination processes can produce parts from a wide range of metals, including titanium, steel, and aluminum. Metal alloys and metal matrix composites (MMCs) can also be used in sheet lamination.

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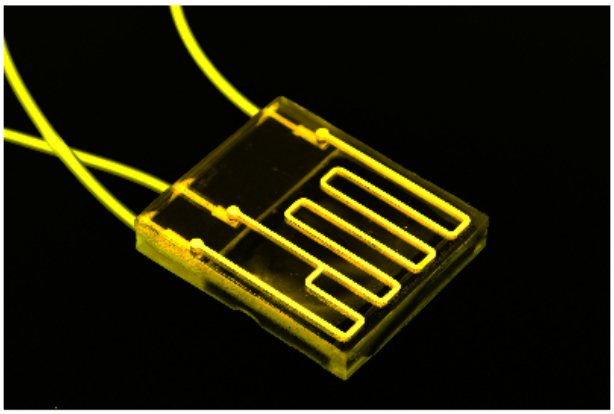
Sheet Lamination Applications

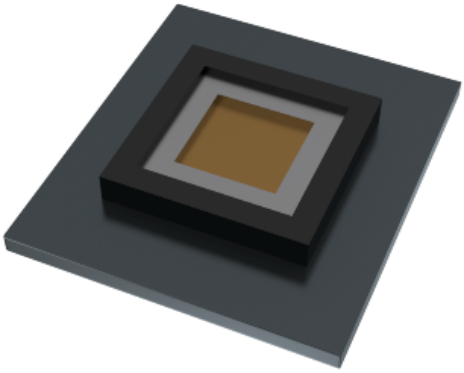
Drone

Like most AM processes, sheet lamination is commonly used to build parts for aerospace components. Sheet lamination processes can print a wide range of aerospace parts used for drones, aircraft, or space exploration, including engine components, wings, and other exterior parts. They are also used to build housings, molds, and other general components for aerospace assembly.



Microfluidic Device

Sheet lamination processes are often used to create ceramic parts for medical applications. Ceramic medical components include bone structures, dental implants, and microfluidic devices. Microfluidic devices can transfer and digitally monitor small amounts of fluid in the human body. These devices are often used to diagnosis illnesses but can also be used in artificial organs.



Multi-Material Part

Sheet lamination can create parts made of different materials or different colors of the same material. In addition, sheet lamination can use a greater number of different materials in a single part more efficiently than most other AM processes. This is because a sheet lamination machine feeds material for a part one sheet at a time, and one roll of material can quickly be removed and replaced with another roll of material.

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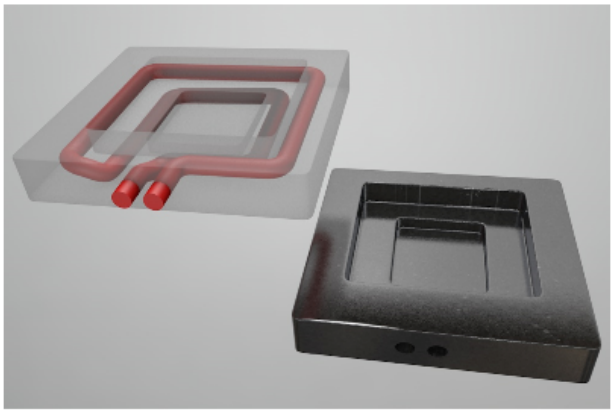
material.



Internal Cooling Channels

Because it is a hybrid manufacturing process, sheet lamination is often used to create complex internal features, like internal channels for enhanced fluid flow or conformal cooling. In hybrid AM processes, subtractive manufacturing tools can machine areas of a part throughout the build process as the part is constructed. This allows manufacturers to finish internal surfaces or create features in areas that would be inaccessible using traditional manufacturing methods. As a result, sheet lamination and other hybrid processes can create more complex parts in a single build process.

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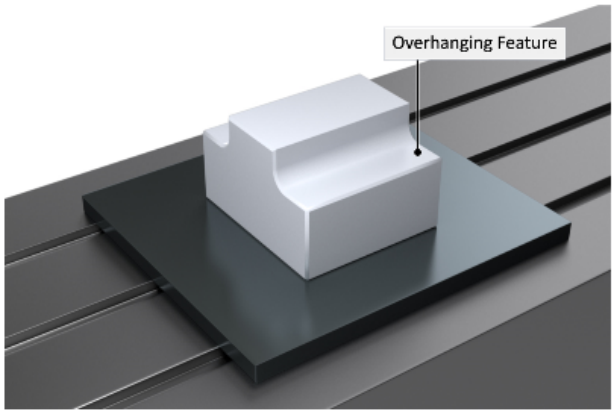
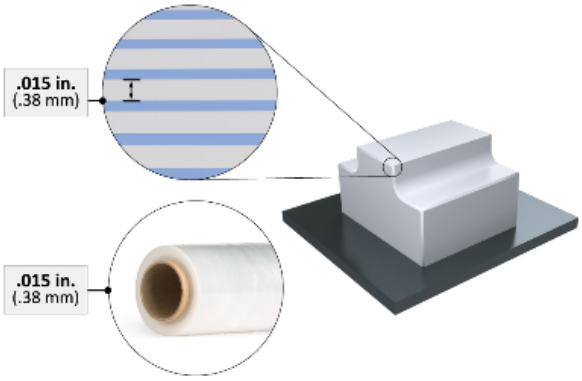
DFAM Considerations for Sheet Lamination



Sheet lamination does not require support structures.

Sheet lamination uses cheaper materials than other AM processes and often prints parts faster, reducing both cost and production time. Since it builds parts by stacking sheets of material, sheet lamination does not require support structures to build overhanging features on a part. Some sheet lamination methods use low temperatures, which is advantageous because intense heating and cooling can change a material's microstructure and properties. Temperature changes can alter part dimensions and compromise part quality. The ability to use multiple materials in a single build efficiently is also a significant advantage of sheet lamination.



Layer thickness depends on material sheet thickness.

One significant drawback of sheet lamination is that the thickness of each part layer depends on the thickness of the material sheet. This means that layer thickness can only be adjusted by using a sheet of the same material with a different thickness. Designers must consider this when building CAD models for sheet lamination. Many sheet lamination methods also create more waste than other AM processes because the excess sheet material not used in the part is simply removed. Additionally, some basic sheet lamination methods cannot create parts with hollow recesses as easily as more advanced AM processes.

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Lesson: 5/10

Review: DFAM for Sheet Lamination

1. Testing is the last phase in the design for additive manufacturing process. **FALSE**
2. Sheet lamination is often considered a hybrid manufacturing process. **TRUE**
3. Sheet lamination can produce microfluidic devices from ceramic materials. **TRUE**
4. Sheet lamination is highly efficient for building a single part that has layers with multiple thickness measurements. **FALSE**
5. Sheet lamination is highly efficient for building a single part made of multiple materials. **TRUE**
6. Sheet lamination methods can use all material types except for ceramic and composite materials. **FALSE**

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Lesson: 6/10

Sheet Lamination Prototyping Methods
The term "sheet lamination" originally referred to Laminated Object Manufacturing (LOM), developed by Cubic Technologies (formerly Helisys). In traditional LOM, paper sheets from a feedstock roll are stacked onto a build platform. A laser or metal blade cuts the shapes of the part layers into each sheet. A heated roller applies pressure, low heat, and an adhesive, often polyurethane glue, to the cut sheet to bond it to the previous sheet.

Like LOM, Selective Deposition Lamination (SDL), a similar process, also cuts and bonds part layer shapes from material sheets laid out successively on a build platform. However, SDL only applies adhesive to specific areas on the sheet that form the layers of the actual part, rather than to the entire sheet.

Plastic Sheet Lamination (PSL) is similar to traditional LOM but uses polymer material sheets. Melting temperatures used in PSL to bond part layers are typically higher than in traditional LOM methods that use paper material. LOM, SDL, and PSL are useful for creating non-functional prototypes but are not used to create end-use parts.

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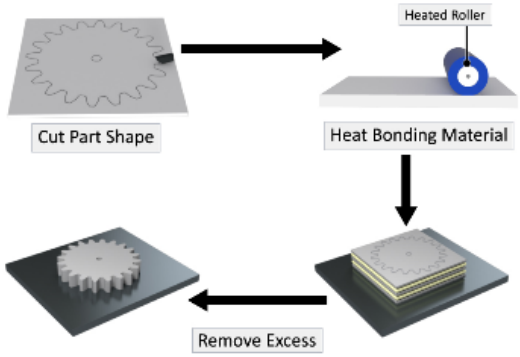
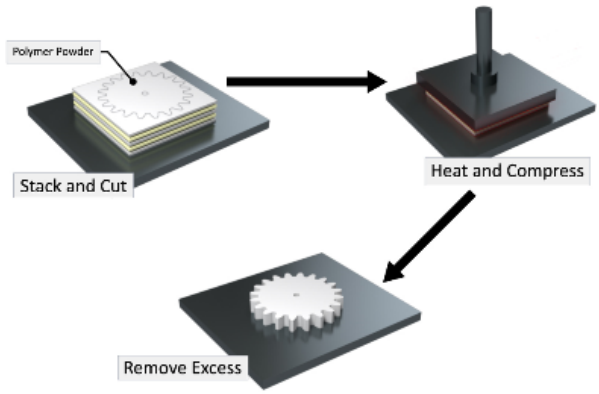
Sheet Lamination Production Methods



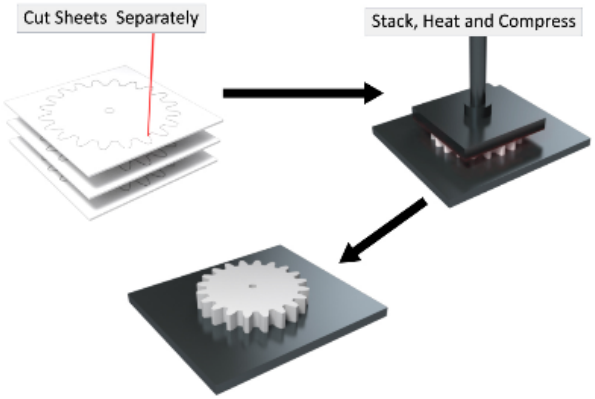
Selective Lamination Composite Object Manufacturing

Selective Lamination Composite Object Manufacturing (SLCOM) is a proprietary sheet lamination method developed by EnvisionTEC, Inc. SLCOM can create very large composite parts made from woven fibers of metal, ceramic, and polymer materials that contain a thermoplastic binding material. During SLCOM, an inkjet printer head prints wax and an adhesive in the shape of the part layer onto the material sheet. Next, a blade cuts out the part shape. A heated roller applies pressure to the part layer that melts the thermoplastic, binding the sheet to the previous layer before a new material sheet is stacked on top. This process repeats until the part is complete.



Composite-Based Additive Manufacturing

Composite-Based Additive Manufacturing (CBAM), developed by Impossible Objects Inc., can also print composites made from different combinations of ceramic, metal, and polymer fibers. Like SLCOM, CBAM uses an inkjet to print part features on material sheets. However, instead of an adhesive, CBAM applies a layer of powdered polymer material that sticks to the ink. Once all layers are complete, CBAM uses heat and compression to melt the polymer powder and fuse the layers into a solid part. CBAM can build parts from a wide range of materials, including polyetheretherketone (PEEK) and nylon, but may require post-processing to remove excess powder.



Computer-Aided Manufacturing of Laminated Engineering Materials

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Computer-Aided Manufacturing of Laminated Engineering Materials (CAM-LEM), developed by CAM-LEM, Inc., can also build ceramic and metal parts, as well as composite parts. Unlike SLCOM and CBAM, the CAM-LEM process cuts each layer of the part separately with a laser before stacking and bonding all the layers together. This method allows excess material to be removed throughout the build process instead of when the part is complete. It also improves the accuracy and bonding strength of each separate layer throughout the build process. Each separately cut layer is lined with an adhesive. Finally, a heat treatment process melts away the adhesive and fuses the sliced layers into a solid part.

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Ultrasonic Additive Manufacturing
The most common sheet lamination method for producing metal parts is Ultrasonic Additive Manufacturing (UAM), or Ultrasonic Consolidation, which was developed by Fabrisonic. UAM uses ultrasonic waves transmitted through a sonotrode to bond sheets of metal tape together. The sheets of metal tape used in UAM are narrower than the material sheets used in other sheet lamination methods like SLCOM or CBAM.

During UAM, the vibration of the waves creates friction that bonds the metal tape to the surface of a build plate or an existing part when pressure and a small amount of heat are applied. UAM machines use metal cutting tools, such as end mills and drills, to machine the features and dimensions of each layer to CAD model specifications. Since the ultrasonic waves can bond material together using friction between the materials' surfaces, UAM does not require intense heat to bond metal surfaces together.

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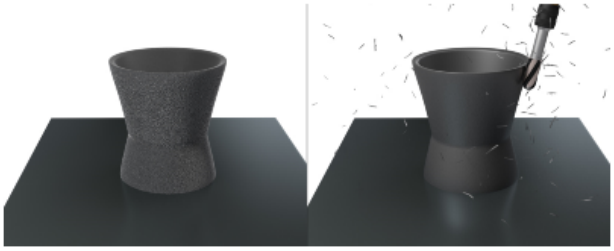
Post-Processing for Sheet Lamination



Sheet lamination uses subtractive manufacturing methods throughout the build process but may also require subtractive manufacturing after the build is complete. One reason subtractive post-processes are needed is because sheet lamination builds parts at lower resolution than other AM methods. This means manufacturers must build a part slightly larger than specified in the CAD model, then machine it down to the measurement specifications. Once machined to spec, parts made with sheet lamination also require abrasive finishing tools to produce a quality surface and bring parts into tolerance. Abrasive finishing processes use tools such as power sanders and grinders.

In addition, sheet lamination processes that use intense heat typically require heat treatment to control cooling rates since rapid temperature changes alter a part's microstructure, which could alter mechanical properties and damage or ruin a part. Sheet lamination processes that require heat treatment include CBAM, CAM-LEM, and PSL.

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Lesson: 10/10

Review: Sheet Lamination Processes

- **Laminated Object Manufacturing**: The original sheet lamination process that uses paper material sheet.
- **Plastic Sheet Lamination**: A sheet lamination process that cuts and melts polymer sheets to make non-functional prototypes.
- **Ultrasonic Additive Manufacturing**: The most common sheet lamination method used for metal parts.
- **Computer-Aided Manufacturing of Laminated Engineering Materials**: A sheet lamination process that cuts part layers separately before stacking and bonding them.
- **Selective Lamination Composite Object Manufacturing**: A sheet lamination process that builds large composite parts from woven fibers.
- **Composite-Based Additive Manufacturing**: A sheet lamination process that may require post-processing to remove excess powder.

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Class Vocabulary

Abrasive Finishing	The use of an abrasive, such as sandpaper, to polish and smooth the surface of a part. Abrasive finishing is commonly used to improve the surface finish of an additively manufactured part.
Additive Manufacturing	AM. The process of joining or solidifying materials to make an object from a three-dimensional computer model. Additive manufacturing methods typically build up layers of material to create an object.
Adhesive	A substance, such as glue, that bonds two surfaces together. Adhesives are used in some sheet lamination operations to fuse part layers.
Aerospace	The industry concerned with the research, design, manufacture, and operation of air and space craft. Aerospace products include airplanes, helicopters, rockets, missiles, satellites, space capsules, space planes, and related systems.
Aluminum	A lightweight metal that is silvery white in color. Aluminum resists corrosion and is a good conductor of electrical and thermal energy.
AM	Additive manufacturing. The process of joining or solidifying materials to make an object from a three-dimensional computer model. AM methods typically build up layers of material to create an object.
Build Platform	The flat surface on which a part is additively manufactured. The build platform on a sheet lamination machine lowers as each layer of the part is completed.
CAD	Computer-aided design. A method of designing two- and three-dimensional objects using computers and software. CAD is most often used to create part models for production.
CAM-LEM	Computer-Aided Manufacturing of Laminated Engineering Materials. A sheet lamination process that cuts each layer of a part separately, then stacks and bonds them to build a final part. CAM-LEM is a proprietary technology of CAM-LEM, Inc.
Carbon Fiber	A material made from slender, thread-like strands of carbon, a strong nonmetallic element. Carbon fiber is a rigid material with good tensile strength, chemical resistance, and temperature tolerance.
CBAM	Composite-Based Additive Manufacturing. A sheet lamination process that stacks and cuts composite sheets of material into layers successively and uses heat, pressure, and binding material to form a solid part. CBAM is a proprietary technology of Impossible Objects, Inc.
Cement	A particulate composite usually composed of clay and limestone. Cement can be used to make mortar, concrete, or other substances depending on how it is mixed.
Ceramic	A hard, brittle material that can withstand high temperatures and resist corrosion. Ceramics, which include metal oxides, nitrides, and glasses, are sometimes used as build material in additive manufacturing.
Ceramic Matrix Composites	CMCs. A composite made from a ceramic matrix and reinforcement materials. Ceramic matrix composites are stiff, lightweight, and can withstand extremely high temperatures.
Composite	A material composed of two or more unlike materials that are bonded together without losing their individual properties or characteristics. Composites consist of a matrix material and a reinforcement material.
Composite-Based Additive Manufacturing	CBAM. A sheet lamination process that stacks and cuts composite sheets of material into layers successively and uses heat, pressure, and powdered material bond layers into a solid part. Composite-Based Additive Manufacturing is a proprietary technology of Impossible Objects, Inc.
Computer-Aided Design	CAD. A method of designing two- and three-dimensional objects using computers and software. Computer-aided design is most often used to create part models for production.
Conceptualization	To form an idea about a possible part or manufactured good. Conceptualization is the initial

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stage of design for manufacturing (DFM) and design for additive manufacturing (DFAM).

Conformal Cooling A feature in a part or component in which internal cooling channel configurations closely correspond to the shape of a part. Conformal cooling allows a part to cool more uniformly, improving heat management, which can improve overall part quality.

Corrosion A process by which a material gradually degrades or wears away. Corrosion typically occurs when a material is exposed to atmosphere, moisture, or other substances.

Design For Additive Manufacturing DFAM. The methodology of planning, testing, and creating an additively manufactured part that functions optimally. Design for additive manufacturing allows engineers to mostly focus on part functionality.

Designing The process of creating the actual specifications for a potential part. Designing involves creating the prints and prototypes for a part.

Drills A multi-point cutting tool used to create a round hole in a workpiece surface. Drills are common tools for holemaking operations.

Drones A compact mobile robot that can be guided through programming or the use of a remote control. Drones used in aerospace and military applications often fly, swim, or both.

Ductile A material's ability to be drawn, stretched, or formed without breaking. Ductile materials bend easily but resist fracture.

End Mills A milling cutter that performs a mix of peripheral and face milling with its bottom and side cutting edges. End mills can be used to machine grooves, slots, circular slots, and pockets, among other features.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing include medical implants, custom dental devices, and camera equipment.

Feedstock Raw material that is used to make manufactured parts. Feedstock for additive manufacturing methods includes filaments, pellets, wire, powder, and material sheets.

Feedstock Roll The cylindrical mechanism on a sheet lamination machine that holds and dispenses thin sheets of build material. Feedstock rolls often hold rolls of paper, polymer, or metal.

Grinders A machine that uses an abrasive to remove material from the surface of a workpiece. Grinders commonly use abrasive grains bonded into a wheel shape to remove small amounts of material in order to improve surface finish.

Heat Treatment A controlled heating and cooling process used to change the structure of a material and alter its physical and mechanical properties. Heat treatments are sometimes used to improve the properties of additively manufactured parts.

Housings A protective cover designed to contain or support a component. Housings may be used to enclose electric motors, gear drives, or other components.

Hybrid Manufacturing A manufacturing approach that produces parts by combining multiple manufacturing processes in a single digital workflow or on a single machine. Hybrid manufacturing may describe a variety of manufacturing process combinations but is most often associated with combining additive and subtractive processes.

Inkjet A device on a traditional printer that emits ink. Some additive manufacturing machines use an inkjet to create the shape of each part layer.

Laminated Object Manufacturing LOM. A sheet lamination process that cuts part features into material sheets using a laser or metal cutting tool and applies an adhesive to the full surface of the material sheets to bond them together. Laminated Object Manufacturing most frequently uses paper material sheets but can also use thermoplastic and composite material sheets.

Laser An intense beam of coherent, collimated, monochromatic light that can be precisely aimed and controlled. Laser can be used for cutting, welding, and other manufacturing processes.

Layer Thickness The amount of material deposited or solidified into a single layer in one pass of an additive manufacturing (AM) process. Layer thickness helps determine build rates, with thicker layers resulting in fast build rates but less accurate parts.

LOM Laminated Object Manufacturing. A sheet lamination process that uses an adhesive to bond successive layers of material together to build a final part. LOM most frequently uses paper

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as build material but can also use thermoplastic and composite sheets.

Metal A naturally occurring material with high electric and thermal conductivity, luster, density, and strength. Metals include copper, iron, nickel, and lead.

Metal Alloys A material consisting of a mix of two or more elements, one of which must be a metal. Metal alloys include cast iron and different types of steels.

Metal Matrix Composites MMCs. A composite made from a metallic matrix and high-performance reinforcements. Metal matrix composites have high levels of strength, stiffness, and relatively high resistance to heat.

Microfluidic Devices A small ceramic component that transfers and digitally processes fluids that can be measured in micrometers. Microfluidic devices can be built using ceramic additive manufacturing methods and are often used in medical applications.

Microstructure The shape and alignment of microscopic components in a material. Microstructure influences the properties of a material.

Molds A hollow cavity that holds heated liquid and imparts its shape on the material as it cools. Molds with complex geometry can be created using sheet lamination and other additive manufacturing processes.

Non-Functional Prototypes A physical model that can represent the scale and appearance of the part or component does not preform functional uses of the part. Non-functional prototypes can be created with sheet lamination and other AM methods.

Nylon An artificial material made from polymers. Nylon rope is very strong, flexible, and resilient.

Overhanging Features Structures on a part that extend without support beneath them. Overhanging features beyond a 45-degree angle typically require support structures in most additive processes.

Planning Assessing the ways to best create a conceptualized part. Planning is a phase in the design process that involves initial considerations of the part design and practical concerns related to the manufacturing process.

Plastic Sheet Lamination PSL. A sheet lamination process that uses heat to bond successive layers of polymer materials together without applying an additional adhesive. Plastic Sheet Lamination is typically used for rapid prototyping.

Polyetheretherketone PEEK. An organic thermoplastic polymer used to fabricate items for high-performance applications. Polyetheretherketone is extensively used in the aerospace, automotive, teletronic, and chemical process industries.

Polymer A lightweight material that generally has high corrosion resistance, a high strength-to-weight ratio, and a low melting point. Polymers, or plastics, are some of the most common manufacturing materials.

Polypropylene A thermoplastic polymer used as a resin for traditional composites. Polypropylene is economical, versatile, and very lightweight.

Polystyrene A thermoplastic that can be molded or foamed. Polystyrene is often used in insulation.

Polyurethane A thermoset polymer that is versatile and wear-resistant. Polyurethane is used in resins for adhesives, elastomers, and fillers.

Post-Processing A group of procedures that are used to clean, improve, or otherwise finish a part for use by a manufacturer or consumer. Post-processing for additively manufactured parts includes abrasive finishing, heat treatment, and painting.

Powdered Turned into a collection of small, uniform, and separate particles. Powdered materials are fused together into solid part in powder bed fusion (PBF) and other additive manufacturing (AM) processes.

Power Sanders A handheld electric tool that uses an abrasive to smooth the surface of an object. Power sanders are often used in post-processing for additive manufacturing.

Production The manufacturing of a finished part for delivery to a customer. Production is the final stage of design for manufacturing (DFM).

Prototype A preliminary model of a part used to evaluate the look and performance of a design. Prototypes are used to determine the specifications for the final part.

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PSL Plastic Sheet Lamination. A sheet lamination process that uses heat to bond successive layers of polymer materials together without applying an additional adhesive. PSL is typically used for rapid prototyping.

Rapid Prototyping A product development technique that uses additive manufacturing methods to create a part model. Rapid prototyping allows manufacturers to quickly create functional part models in a short time period, reducing lead time.

Resolution The fineness of detail on a part specified in a computer-aided design (CAD) model. Resolution helps determine the surface finish of an additively manufactured part, with greater resolution leading to better surface finish.

Selective Deposition Lamination SDL. A sheet lamination process that cuts part features into material sheets using a laser or metal cutting tool and applies an adhesive in select areas of the material sheets to bond them together. Selective Deposition Lamination most frequently uses paper and only bonds areas of the sheets that will form the part rather than bonding entire material sheets together.

Sheet Lamination An additive manufacturing method that forms a part layer by layer by bonding sheets of material together using an adhesive, heat, pressure, or ultrasonic vibrations. Sheet lamination methods include Laminated Object Manufacturing, Ultrasonic Additive Manufacturing, and other processes.

SLCOM Selective Lamination Composite Object Manufacturing. A sheet lamination process that cuts sheets of material into layers of a part separately before stacking and bonding the layers into a solid object. SLCOM is a proprietary technology of EnvisionTEC, Inc.

Sonotrode A tool on an ultrasonic welding or additive manufacturing (AM) machine that transmits ultrasonic vibrations. The sonotrode on a UAM machine helps create the friction to bond surfaces together.

Specifications A description of the essential physical and technical properties of a finished part. Specifications outline important information including finished part dimensions and how the part must respond to forces acting upon it.

Specifications A detailed plan for a part or object that includes dimensions and other precise descriptions of its manufacturing requirements. Specifications are typically created using computer-aided design (CAD) software.

Steel A metal made primarily of iron and carbon that is alloyed with small amounts of other elements. Steels often contain elements such as manganese, phosphorous, sulfur, and silicon to enhance various properties of the metal.

Subtractive Manufacturing A manufacturing process that machines a piece of raw material into a desired final shape through a controlled material removal process. Subtractive manufacturing methods are fast but create a large amount of waste.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Testing The examination of a part to ensure that it performs its intended function and that it can be satisfactorily manufactured. Testing is a design step that indicates whether the part needs additional planning or if it is ready for production.

Thermoplastics A type of polymer that can be repeatedly heated, shaped, and cooled. Thermoplastics are sometimes used in additive manufacturing processes, including sheet lamination.

Thermosets A type of plastic, also known as thermosetting plastics, that cannot be reheated or re-formed after their initial heating. Thermosets are sometimes used in additive manufacturing processes, including sheet lamination.

Titanium A nonferrous metal that is lightweight and corrosion resistant and has a high strength-to-weight ratio. Titanium and its alloys are often used in the aerospace industry.

Tolerance An unwanted but acceptable deviation from a desired dimension. Tolerances indicate the allowable difference between a physical feature and its intended design.

Ultrasonic A frequency above the range of human hearing. Ultrasonic vibrations are used to bond sheets of metal in Ultrasonic Additive Manufacturing (UAM).

Ultrasonic Additive Manufacturing UAM. A sheet lamination process that uses high-frequency vibrations to bond successive layers of material together to build a final part. Ultrasonic Additive Manufacturing, also known

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layers of material together to build a final part. Ultrasonic Additive Manufacturing, also known as Ultrasonic Consolidation, most frequently uses sheets of metal foil or tape as build material.

Ultrasonic Consolidation A sheet lamination process that uses high-frequency vibrations to bond successive layers of material together to build a final part. Ultrasonic Consolidation, also known as Ultrasonic Additive Manufacturing (UAM), most frequently uses sheets of metal foil or tape as build material.

Woven Fibers Thin strands of material stitched tightly together. Woven fibers may improve strength and other mechanincal properties of the material compared to unwoven fibers.

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